Question 1: Containerize a sample application using Docker: write a Dockerfile, build the image, and run the container. Document the image layers and container behavior.

Executed on Ubuntu 24.04 LTS (WSL2) by Meenakshi Rakesh. Repository: https://github.com/meenakshi16rp/Devops-lab-2

1. The Dockerfile

Dockerfile (committed at the repository root)

```
# ENSP461 Lab 2 - sample application image
FROM python:3.12-slim

WORKDIR /app

COPY app/requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

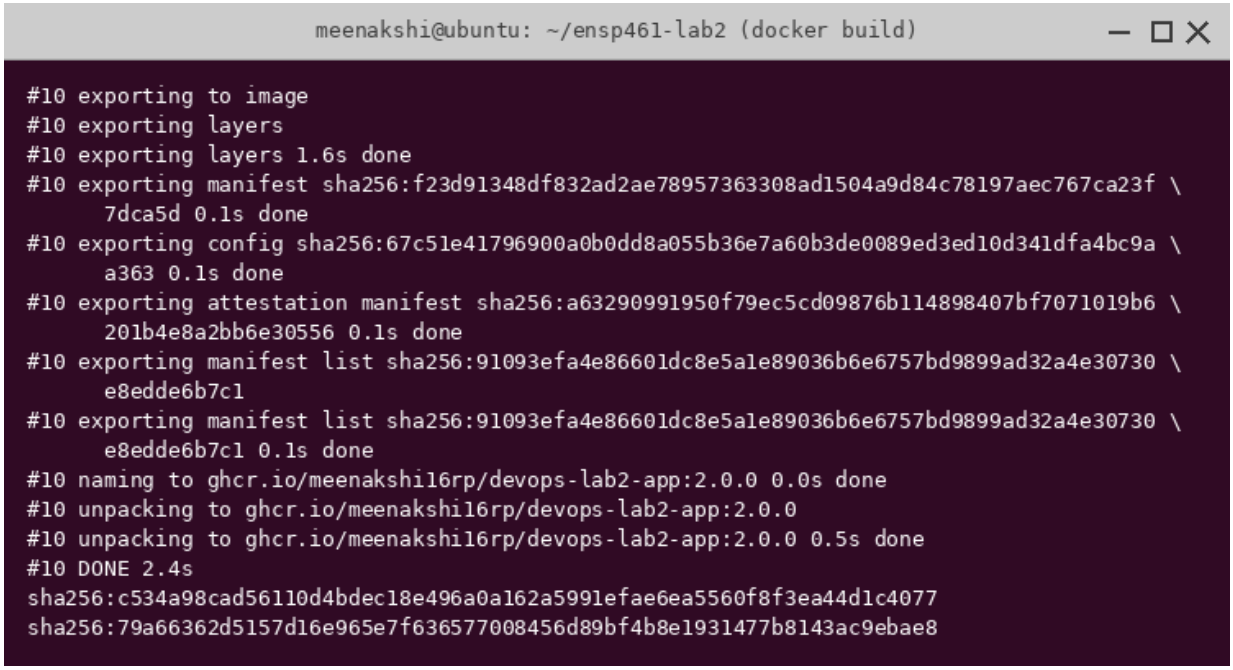
COPY app/ .

EXPOSE 5000
ENV APP_VERSION=2.0.0

CMD ["python3", "app.py"]
```

- Base image python:3.12-slim keeps the image small (207 MB disk usage, 50.3 MB content).
- requirements.txt is copied and installed before the source, so dependency layers stay cached when only the app code changes.

2. Building the image




```
meenakshi@ubuntu:~$ docker images ghcr.io/meenakshi16rp/devops-lab2-app
REPOSITORY                              TAG       SIZE
ghcr.io/meenakshi16rp/devops-lab2-app   2.0.0     207MB
```

3. Running the container and verifying behavior

```
meenakshi@ubuntu: ~/ensp461-lab2 (docker run)

meenakshi@ubuntu:~$ docker inspect lab2-web --format ...
Image: ghcr.io/meenakshi16rp/devops-lab2-app:2.0.0 | State: running | Bridge IP: \
      172.17.0.2

meenakshi@ubuntu:~$ docker logs lab2-web
 * Serving Flask app 'app'
 * Debug mode: off
□[31m□[1mWARNING: This is a development server. Do not use it in a production \
      deployment. Use a production WSGI server instead.□[0m
```

4. Image layers

```
meenakshi@ubuntu: ~/ensp461-lab2 (docker history)

<missing>      7 days ago      CMD ["python3"]                                 0B       \
      buildkit.dockerfile.v0
<missing>      7 days ago      RUN /bin/sh -c set -eux;  for src in idle3 p…   16.4kB   \
      buildkit.dockerfile.v0
<missing>      7 days ago      RUN /bin/sh -c set -eux;   savedAptMark="$(a…   41.4MB   \
      buildkit.dockerfile.v0
<missing>      7 days ago      ENV PYTHON_SHA256=b5796915dc45ff745cfb4e990a…   0B       \
      buildkit.dockerfile.v0
<missing>      7 days ago      ENV PYTHON_VERSION=3.12.16                      0B       \
      buildkit.dockerfile.v0
<missing>      7 days ago      ENV GPG_KEY=7169605F62C751356D054A26A821E680…   0B       \
      buildkit.dockerfile.v0
<missing>      7 days ago      RUN /bin/sh -c set -eux;  apt-get update;  a…   4.93MB   \
      buildkit.dockerfile.v0
<missing>      7 days ago      ENV LANG=C.UTF-8                                0B       \
      buildkit.dockerfile.v0
<missing>      7 days ago      ENV PATH=/usr/local/bin:/usr/local/sbin:/usr…   0B       \
      buildkit.dockerfile.v0
```

- docker history shows each Dockerfile instruction as a layer: WORKDIR, COPY requirements (12.3 kB), pip install (22.6 MB), COPY source (20.5 kB), EXPOSE/ENV/CMD metadata layers.
- Container behavior: the Flask server binds 0.0.0.0:5000, / returns the greeting and version, /health returns UP, and /info reports the container hostname.

5. Observations

- The first run exited silently because app.py had no app.run() entry point - the container exited 0 with no logs; the fix was adding the entry block and rebuilding.
- A missing jsonify import then produced HTTP 500 with a NameError in docker logs - the console log pinpointed both defects quickly.
- Layer caching: rebuilding after changing only app/app.py reused the pip install layer, making the rebuild finish in seconds.